

Name : Qadeer Durrani

Roll no : su91-bsram-f25-040

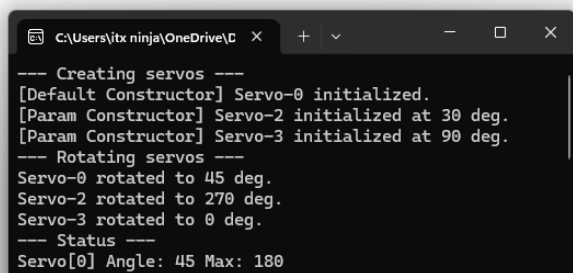
Dept : Robotics and Ai

OOPS

Lab 8

Task 1

```
#include <iostream>
using namespace std;
class ServoMotor {
private:
    int servoID;
    float currentAngle;
    float maxAngle;
public:
    ServoMotor() {
        servoID = 0;
        currentAngle = 0.0f;
        maxAngle = 180.0f;
        cout << "[Default Constructor] Servo-0 initialized." << endl;
    }
    ServoMotor(int id, float angle, float maxAng)
        : servoID(id), currentAngle(angle), maxAngle(maxAng)
    {
        cout << "[Param Constructor] Servo-" << servoID
            << " initialized at " << currentAngle
            << " deg." << endl;
    }
};
```

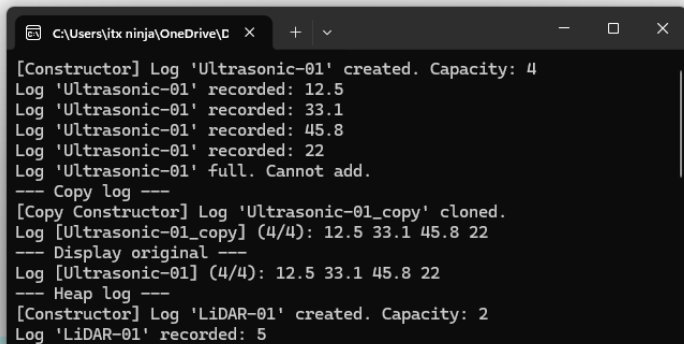


```
C:\Users\jtx ninja\OneDrive\...
--- Creating servos ---
[Default Constructor] Servo-0 initialized.
[Param Constructor] Servo-2 initialized at 30 deg.
[Param Constructor] Servo-3 initialized at 90 deg.
--- Rotating servos ---
Servo-0 rotated to 45 deg.
Servo-2 rotated to 270 deg.
Servo-3 rotated to 0 deg.
--- Status ---
Servo[0] Angle: 45 Max: 180
```

Task 2 :

```
#include <iostream>
using namespace std;
class SensorLog {
private:
    string sensorID;
    float* data;
    int capacity;
    int count;
public:
    SensorLog(string id, int cap)
        : sensorID(id), capacity(cap)
    {
        count = 0;
        data = new float[capacity];
        for(int i = 0; i < capacity; i++)
            data[i] = 0.0f;

        cout << "[Constructor] Log '" << sensorID
            << "' created. Capacity: "
            << capacity << endl;
    }
    SensorLog(const SensorLog& other)
    {
        sensorID = other.sensorID + "_copy";
    }
};
```



```
C:\Users\jtx ninja\OneDrive\...
[Constructor] Log 'Ultrasonic-01' created. Capacity: 4
Log 'Ultrasonic-01' recorded: 12.5
Log 'Ultrasonic-01' recorded: 33.1
Log 'Ultrasonic-01' recorded: 45.8
Log 'Ultrasonic-01' recorded: 22
Log 'Ultrasonic-01' full. Cannot add.
--- Copy log ---
[Copy Constructor] Log 'Ultrasonic-01_copy' cloned.
Log [Ultrasonic-01_copy] (4/4): 12.5 33.1 45.8 22
--- Display original ---
Log [Ultrasonic-01] (4/4): 12.5 33.1 45.8 22
--- Heap log ---
[Constructor] Log 'LiDAR-01' created. Capacity: 2
Log 'LiDAR-01' recorded: 5
```

Home task :

```

#include <iostream>
using namespace std;
class RobotArm {
private:
    string armName;
    int numJoints;
    string* jointNames;
    float* jointAngles;
    float maxReach;
public:
    RobotArm()
    {
        armName = "DefaultArm";
        numJoints = 3;
        maxReach = 50.0f;
        jointNames = new string[numJoints];
        jointAngles = new float[numJoints];
        jointNames[0] = "J1";
        jointNames[1] = "J2";
        jointNames[2] = "J3";
        for(int i = 0; i < numJoints; i++)
            jointAngles[i] = 0;
        cout << "[Default Constructor] "
              << armName
              << " created. Joints: "
              << numJoints << endl;
    }
};

```

```

C:\Users\itx.ninja\OneDrive\...
[Default Constructor] DefaultArm created. Joints: 3
[Param Constructor] Arm-Delta created. Joints: 4
Shoulder set to 45 deg
Elbow set to 90 deg
Wrist set to 30 deg
[Copy Constructor] Arm-Delta_copy cloned. Joints: 4
--- Arm-Delta Status ---
Arm: Arm-Delta | Max Reach: 120 cm
Joint[0] Shoulder : 45 deg
Joint[1] Elbow : 90 deg
Joint[2] Wrist : 30 deg
Joint[3] Gripper : 0 deg
Total angle: 165 deg

```